



Bond return predictability: Evidence from 25 OECD countries

Neluka Devpura^a, Paresh Kumar Narayan^{b,*}, Susan Sunila Sharma^c^a Department of Statistics, University of Sri Jayewardenepura, Sri Lanka^b Monash Business School, Monash University, Melbourne, Australia^c Centre for Financial Econometrics & Department of Finance, Deakin Business School, Deakin University, Melbourne, Australia

ARTICLE INFO

Article history:

Received 1 September 2020

Accepted 24 January 2021

Available online 2 March 2021

Keywords:

Panel predictive regression

Heteroskedasticity

Bond excess returns

Commodities

ABSTRACT

In this paper, we test for bond excess return predictability in 25 Organisation for Economic Co-operation and Development countries, using a set of 12 predictor variables, including macroeconomic, financial, and commodity-based indicators. The novelty of our approach is that we employ both time-series and panel data models that accommodate key data features, namely, persistency, endogeneity, heteroscedasticity, and cross-sectional dependence. We note two key findings. The first is that commodity variables, both oil spot and futures prices, and the world commodity price index predict bond excess returns. Our second finding confirms that macroeconomic variables (the term spread and T-bill yield) are successful predictors of bond excess returns.

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1. Introduction

Bond excess return predictability has attracted the attention of financial economists and investors for many years. Investors are keen on investing in fixed asset classes such as Treasury bonds, because they play a major role in portfolio diversification. Bonds offer a risk-free yield curve and attract bond investors. Thus, specially looking at the emerging markets, Turner (2012) argued that the domestic bond market in local currency reduced currency mismatches and resilience during the Lehman crisis. Park et al. (2019) find domestic bond markets' stabilizing ability of emerging economies in their analysis of the global financial crisis and the taper tantrum. Therefore, bonds as fixed assets serve as a stabilizer of markets, especially during the crises.

Compared to stock return predictability, studies on bond excess return predictability are limited. One strand of literature (Cochrane and Piazzesi, 2005; Thornton and Valente, 2012) documents evidence in favor of bond return predictability when using forward rates or forward spreads as predictors. Cochrane and Piazzesi (2005) find evidence that a linear combination of forward rates has predictive ability. Thornton and Valente (2012) investigate the out-of-sample (OOS) predictability of bond excess returns based on long-term forward rates and conclude that forward rates predict bond excess returns.

A feature of this literature is that only a few studies analyze the predictability of government bond returns using non-US data. These studies constitute a second strand of the literature. Ilmanen (1995), for instance, examines the predictability of government bond returns in six countries (the US, Canada, Japan, Germany, France, and the UK). The author finds that global factors predict returns across countries. Campos, Cortez, and Armada (2003) conduct a study on the predictability of bond returns (non-US data) for the European bond market. They find that the term spread, and January dummy variables predict the European bond market.

The third strand of this literature argues that variables useful in predicting stock returns should also predict bond returns. These studies use dividend yields and interest rates to forecast stock and bond returns. Using term spread, the default spread,

* Corresponding author.

E-mail address: pareshkumar.narayan@monash.edu (P.K. Narayan).

and dividend yield, Fama and French (1989) find that variations in expected returns are common to various asset classes, such as stocks and bonds. Using as many as 132 macroeconomic variables, Ludvigson and Ng (2009) reveal that factors generated from macroeconomic variables predict Treasury bonds. Kinateder and Papavassiliou (2019) show that kurtosis predicts bond returns. Fong and Wu (2020) use technical trading rules to predict sovereign bonds for a large number of markets and show that trading rules successfully predict bond returns particular in Asian markets. Cao, Galvani and Gubellini (2017) investigate the role of firm-specific factors in predicting Canadian bonds and find a role for mean of firm level factors. Narayan et al. (2016) show that in the case of US, equities do influence bond returns.

This paper focuses on this third strand of the literature. We investigate whether bond excess returns are predictable when subjected to a range of macroeconomic, financial, and commodity market-based predictors. Our choice of predictors is motivated by empirical evidence that they predict stock returns. Our approach consists of four steps. First, we use historical monthly data from January 1920 to December 2016. Historical data are useful in identifying predictable relations (e.g., Goyal and Welch, 2008; Narayan and Gupta, 2015; and Devpura, Narayan, and Sharma, 2018).

Second, we consider a comprehensive set of 12 predictor variables. The multitude of predictors allows us to test their relative importance. Specifically, we use five commodity variables: the logarithm of the Goldman Sachs commodity price index (WCPI), the logarithm of the gold spot price (GP), the logarithm of the gold futures price (GFP), the logarithm of the oil spot price (OP), and the logarithm of the oil futures price (OFF). There are five macroeconomic variables: the three-month Treasury bill yield (*T-bill*), to represent the short-term interest rate; the term spread (*TS*); the logarithm of the exchange rate (*ER*); the logarithm of the Consumer Price Index (*CPI*); and the percentage growth rate of industrial production (*IP*). The two financial variables are the logarithm of the stock market index (*Stock*) and the logarithm of the dividend yield (*DY*). Using a set of variables representing different parts of the economic and financial system helps identify any trends in predictability. The inspiration behind our choice of predictors is that these variables are widely used in testing stock return predictability. We discuss and motivate the choice of variables in Section 2.

Third, we consider both time-series and panel data predictive regression models. Given data availability, time-series predictive regression models are commonly used for forecasting. The motivation behind a panel data approach is that it allows us to explore cross-sectional variations in the data in addition to time-series variations, thus offering an additional dimension through which predictability can be explored. Thus, panel data increase the total number of observations. More recently, Westerlund, Karabiyik, and Narayan (2017) have developed a new panel-based predictability model that is both robust and powerful in forecasting stock returns. For panel regression, we form multiple country panels. Our main panel is comprised of 25 Organisation for Economic Co-operation and Development (OECD) countries. For robustness, we consider three additional panel formations: for a G7 country, for an OECD European region, and an OECD high-income country.

Fourth, we use the time-series and panel data predictive regression estimators proposed by Westerlund and Narayan (2015, hereafter WN) and Westerlund et al. (2017), respectively. These estimators are popular because they are inspired by such data features as persistency, endogeneity, heteroscedasticity, and cross-sectional dependence, which need to be modeled to obtain unbiased estimates.

Our approaches contribute four new sets of results. Our first main finding relates to the predictors. Using panel regression, we show that the commodity variables, namely, *OP* and *OFF*—which, until now, have not been modeled as predictors of bond excess returns—turn out to be successful predictors. We also find *WCPI* predicts bond excess returns in three out of four panels. We discuss the motivation behind this type of commodity and bond return relation (see Section 2). Among the five macroeconomic predictors, *TS* consistently shows predictive ability.¹ Regarding the time-series method, *TS* shows both in-sample and OOS predictability, whereas, for panel regressions, all four panel formations show that *TS* predicts bond excess returns. The next most significant predictor is *T-bill*. In panel regressions, for three out of four panels, we find that *T-bill* predicts bond excess returns. From all four panels, we conclude that our finding of bond excess return predictability, particularly through the commodities channel (three of five commodity predictors), holds remarkably well.

In addition, we undertake an extensive OOS test for bond excess return predictability. In both time-series and panel regression settings, we find limited in-sample evidence but strong OOS evidence of predictability. Finally, with respect to the economic significance of the predictability relation, we utilize a power utility function to explore the economic significance of our statistical results. In time-series tests, maximum utility (highest number of positive gains) is recorded by the predictor *GFP*: it shows 11 out of 23 countries with positive utility. The next four predictors with positive utility are *GP*, *CPI*, *TS*, and *T-bill*. For panel-based utility, we find that the utility gains for all 12 predictor variables are positive.

Our approaches and results contribute in several ways to the bond predictability literature. Our first contribution relates to the methodologies on bond return predictability. Gargano, Pettenuzzo, and Timmermann (2017), for instance, consider both univariate and multivariate prediction models; Ludvigson and Ng (2009) use principal component analysis to estimate the common factors from 132 predictors; and Guidolin, Hyde, McMillan, and Ono (2009) consider nonlinear regression models. Ludvigson and Ng (2009) and Gargano et al. (2017) find that macro factors predict bond returns, whereas Guidolin et al. (2009) find that nonlinear models improve forecast performance. We use both time-series and panel data models to control for features such as predictor persistency, endogeneity, heteroscedasticity, and cross-sectional dependence, and we find superior OOS forecast performance compared to in-sample results from both the time-series and panel models. We also document that statistical evidence of predictability translates into meaningful utility gains for investors.

¹ See for instance, Fama and French (1989) and Ilmanen (1995).

Our second contribution relates to the international bond market. That is, it is reasonable to suppose that countries that share many characteristics are more likely to have similar predictability patterns than those that do not. We consider 25 OECD countries, including the US. For example, [Ilmanen \(1995\)](#) examines bond return predictability for six countries and [Guidolin et al. \(2009\)](#) examine bond return predictability for G7 countries. We add to this literature an analysis based on a larger number of countries, i.e., 25 OECD countries. We also group countries into four panels and we analyze each country's bond predictability separately (time-series). Compared to the literature, we use a substantially higher number of countries (25) and predictor variables (12). We therefore take a rigorous approach and contribute by showing that (a) the commodity variables *OPF*, *OP*, and *WCPI* predict bond excess returns, which is a new finding for bond excess return predictability; (b) two macroeconomic variables, *TS* and *T-bill*, predict bond excess returns; and (c) these predictors generate gains in OOS forecasts compared to the constant-return benchmark model.

Our third contribution relates to the role played by the commodity market in influencing the bond market. We find evidence of bond excess return predictability through the energy variables *OPF* and *OP*, as well as the *WCPI* commodity index. Similarly, several studies show that stocks can be predicted using commodities, implying that bonds can also be predicted by commodities, given the strong empirical evidence that bonds and stocks co-move (e.g., [Connolly, Stivers, and Sun, 2005](#); [Cappiello, Engle, and Shepperd, 2006](#)). Finding comovement between stocks and bonds, as this literature documents, does not, however, imply the predictability of bond returns through energy variables. Our contribution is our formulation of the hypothesis that energy variables predict bond returns and its empirical testing. We show strong statistical and economic significance of the role played by energy variables in predicting bond returns.

The remainder of this paper proceeds as follows. In [Section 2](#), we discuss the related literature, along with our hypotheses and motivations for testing them. In [Section 3](#), we explain our bond excess return calculation and methodology. [Section 4](#) presents our empirical findings, including both time-series and panel data models and in-sample and OOS test-based results. [Section 5](#) assesses the economic significance of bond return predictability for a risk-averse investor with a risky bond and a risk-free asset. In [Section 6](#), we present robustness checks, and in [Section 7](#) we set forth our concluding remarks.

2. Literature review and hypotheses

Stock return predictability is a popular research subject. Studies use a large number of predictors ([Narayan and Bannigidadmath, 2015](#)) to forecast stock returns. This predictability analysis has focused on four specific asset classes: bonds ([Cochrane and Piazzesi, 2005](#)), stocks ([Goyal and Welch, 2008](#)), commodities ([Hong and Yogo, 2012](#)), and currencies ([Meese and Rogoff, 1983](#)).

Inspired by this literature, we consider 12 commonly used stock return predictors as predictors of bond returns: *GP*, *GFP*, *OP*, *OPF*, *WCPI*, *CPI*, *DY*, *ER*, *IP*, *Stock*, *TS*, and *T-bill*. Our motivation for using these predictors has its roots in empirical evidence of the relation between stocks and bonds. Many studies capture this relation by investigating correlations between stocks and bonds, as discussed below; see [Connolly et al. \(2005\)](#), and [Li \(2002\)](#).

2.1. Motivating commodity variables

Inspired by the notion of open interest, we use four commodity variables relating to oil and gold—the gold spot price, the gold futures price, the oil spot price, and the oil futures price—because these two commodities have been instrumental in influencing asset prices ([Baur and Lucey, 2010](#)).

In addition to oil and gold commodity variables, we consider an aggregate measure of commodity prices, *WCPI*, as a predictor. This allows us to test the robustness of the role of the commodity markets in predicting bond returns. Among these five commodity variables, oil is by far the most extensively studied in the stock return literature (e.g., [Narayan and Sharma, 2011](#); [Narayan and Gupta, 2015](#)).

The importance of gold as a hedging asset or safe haven asset is studied by [Baur and Lucey \(2010\)](#). They find that, while gold is a safe haven for UK and US stocks, this is not the case with respect to the bond market. Finally, we consider the GSCI (which comprises 24 commodities) as a proxy for our predictor, *WCPI*, to detect any ability to predict bond excess returns. In addition, [Jacobson et al. \(2016\)](#) show the predictive ability of GSCI industrial returns for bond returns.

Accordingly, our first hypothesis is as follows:

H1: Commodity variables can potentially predict bond risk premiums (bond excess returns).

2.2. Motivating macroeconomic factors

Our choice of macroeconomic variables as predictors is inspired by the bond return forecasting literature. We consider the common variables in this literature: *TS*, the *T-bill*, *ER*, *CPI*, and *IP*. A number of studies have investigated the relevance of these predictors to forecasting bond excess returns. Among these predictors, *TS* is the most commonly used predictor of bond returns. [Fama and French \(1989\)](#) and [Ilmanen \(1995\)](#) show that *TS* predicts bond excess returns; see also [Chow et al. \(1997\)](#) and [Ilmanen \(1995\)](#) for the relevance of exchange rate (see, [Wang and Bilson, 2017](#)) and T-bill rate, respectively. Very few studies consider the *IP* variable as a predictor of bond return predictability; see [Ludvigson and Ng \(2009\)](#) and [Aslanidis](#)

and Christiansen (2014). For instance, Akgiray et al. (2016) identify the main predictor as factors based on real economic activity that measures the industrial and manufacturing production. Finally, CPI news has been used to test bond return predictability. Smirlock (1986), for instance, examines the response of the long-term bond market to CPI (inflation) announcements; Başçı and Ekinici (2005) evidence the inflation risk mpalins bond premium and Essers et al. (2016) find bond market capitalization is negatively related with inflation. This leads to our second hypothesis.

H2: Macroeconomic variables predict bond excess returns.

2.3. Motivating financial variables

We consider two financial variables among the 12 predictors to determine whether they predict bond excess returns. The predictors of interest are the stock index (*Stock*) and *DY*. Fama and French (1993) investigate the common risk factors for stocks and bonds. Moreover, Cappiello et al. (2006) document that stock and bond returns co-move in a time-varying fashion and Lee, Lee and Zhang (2019) evidence co-movement of Chinese Stock and Bonds. Thus, we argue that stock returns influence bond returns, and vice versa, through information spillover among financial markets.

The last financial variable we consider is *DY*. In the stock market context, the hypothesis is that stock prices are low compared to dividends when discount rates and expected returns are high (and vice versa). Fama and French (1989) use *DY* as a predictor of bond returns and find that it does predict bond returns. This leads to our third hypothesis.

H3: Financial variables predict bond excess returns.

3. Data and methodology

3.1. Data

We use monthly data for 25 OECD countries: Australia, Austria, Belgium, Canada, the Czech Republic, Denmark, Finland, France, Germany, Hungary, Ireland, Italy, Japan, Korea, Mexico, the Netherlands, New Zealand, Norway, Poland, Portugal, Spain, Sweden, Switzerland, the UK, and the US. Our data cover the period from January 1920 to December 2016 and were obtained from the Global Financial Database. We provide detailed data descriptions in the Appendix (see Table A.1). We follow Guidolin et al. (2009) for the selection of mnemonics to download data from the Global Financial Database.

Monthly data are common in the literature (e.g., Fama and French, 1989). For our main empirical analysis, we use both a time-series and a panel approach. More specifically, we use a balanced panel approach for which we need to assure that all 25 OECD countries have the same start and end dates. Thus, data for the balanced panel approach cover the period January 2000 to December 2016. The choice of the data time span is dictated by data availability. To compare the results obtained from panel and time-series setups, we use data with the same start and end dates, irrespective of the approach used for the analysis. Therefore, for time-series analysis, we again use data over the same time span (January 2000 to December 2016).

One could dispute the use of such a short time span, when the data for some countries are available since January 1920. There are two reasons for it. First, since we are using a panel approach, this time span allows us to use the cross-sectional dimension (25 countries), to compensate for the relatively short time span. Thus, we are comfortable with a short time T . Second, we reduce the risk of structural breaks by using a recent, short timespan. In addition, our focus on 25 OECD countries allows us to form three additional homogeneous subpanels, namely, OECD high-income countries, OECD European region countries, and G7 countries.

The next concern relates to our choice of short time span for the time-series approach. The reasons mentioned do not justify the use of a short time span in the case of time-series regressions. Therefore, as a robustness check, we conduct our empirical analysis using the available full data set (January 1920 to December 2016). Note that not all the variables for all 25 OECD countries have the same start and end dates, as dictated simply by data availability; an online Appendix has specific details on dates.

Let us now define our variables. The dependent variable is bond excess returns, which we compute following the approach of Engsted, Moller, and Sander (2013). Bond excess returns, r_t , are simply the natural logarithm of the returns of bond indexes minus the natural logarithm of one plus the three-month Treasury bill rate. The three-month Treasury bill rates are obtained from three-month Treasury bill yield, and since the latter is in annualized percentage form, we transform it into monthly series. We collected the data from Global Financial Database (GFD) and the bond excess returns are calculated from 10-year total return government bond index. Additionally, there are 12 predictors: *Stock*, *DY*, *T-bill* (which represents the short-term interest rate), *TS* (which is the difference between the 10-year bond yield and the three-month Treasury bill yield), *ER*, *CPI*, *IP*, *WCPI*, *GP*, *GFP*, *OP*, and *OFF*. Note that not all 12 predictor variables are available for all 25 OECD countries. For example, the predictor variables *IP*, *TS*, and *CPI* are available for 17, 22, and 23 countries, respectively (see Table 1 for more details).

Table 1

Data availability over the period January 2000 to December 2016. This table provides detail on specific predictor availability for each country over the period January 2000 to December 2016. There are 12 predictor variables, namely, gold futures price (*GFP*), gold spot price (*GP*), oil futures price (*OPF*), oil spot price (*OP*), world commodity index (*WCPI*), consumer price index (*CPI*), dividend yield (*DY*), exchange rate (*ER*), industrial production (*IP*), stock market index (*Stock*), term spread (*TS*) and 3-month Treasury bill rate (*T-bill*).

Predictor	No. of Countries	Country List
<i>WCPI, GFP, GP, OPF, OP, DY, ER, Stock, T-bill</i>	25	Australia, Austria, Belgium, Canada, Czech Republic, Denmark, Finland, France, Germany, Hungary, Ireland, Italy, Japan, Korea, Mexico, the Netherlands, New Zealand, Norway, Poland, Portugal, Spain, Sweden, Switzerland, U.K., and U.S.
<i>IP</i>	17	Austria, Belgium, Canada, Denmark, France, Germany, Ireland, Italy, Japan, Korea, Mexico, the Netherlands, Norway, Portugal, Sweden, U.K., and U.S.
<i>TS</i>	22	Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Hungary, Ireland, Italy, Japan, the Netherlands, New Zealand, Norway, Poland, Portugal, Spain, Sweden, Switzerland, U.K., and U.S.
<i>CPI</i>	23	Australia, Austria, Belgium, Canada, Czech Republic, Finland, France, Germany, Hungary, Ireland, Italy, Japan, Korea, the Netherlands, New Zealand, Norway, Poland, Portugal, Spain, Sweden, Switzerland, U.K., and U.S.

3.2. Estimation approach

To examine the null hypothesis of no predictability, we employ both time-series and a panel predictability approach. For the time-series analysis, we adopt WN's predictability model. Our time-series predictability model takes the following form:

$$r_t = \alpha + \beta x_{t-1} + \varepsilon_t \quad (1)$$

where r_t is the bond excess return, x_t is a predictor, and ε_t is the disturbance term. This framework allows us to model the persistency and endogeneity of the predictor variable and any autoregressive conditional heteroskedasticity (ARCH) effects. Since we use 12 predictor variables, we estimate the model for each OECD country, subject to data availability. We refrain from discussing the WN approach in further detail, because this time-series predictability model is well known in the literature and has been discussed repeatedly (e.g., Sharma, 2016; Devpura et al., 2018).

In addition to the time-series predictive regression, we employ the panel predictive regression model proposed by Westerlund et al. (2017). The data-generating process is written as

$$r_{i,t} = \alpha_i + \beta_i x_{i,t-1} + u_{i,t} \quad (2)$$

$$u_{i,t} = \lambda_i f_t + \varepsilon_{i,t} \quad (3)$$

where $r_{i,t}$ is the bond excess return, α_i is the intercept term that varies for each country, $x_{i,t}$ denotes an $m \times 1$ vector of predictors, f_t is a common factor and is estimated as demeaned bond excess returns, λ_i is the associated factor loading, and $\varepsilon_{i,t}$ is an idiosyncratic error term.

Before we move on to the next equations, let's define *forwards* and *backwards* recursively demeaned values for some generic variable a_t , $a_t^* = a_t - (T - t + 1)^{-1} \sum_{n=t}^T a_n$ and $a_t^{**} = a_t - t^{-1} \sum_{n=1}^t a_n$ respectively. Let, $M_A = I_{T-1} - A(A'A)^{-1}A'$ for any matrix A with $(T - 1)$ rows. And $\bar{a} = N^{-1} \sum_{i=1}^N a_i$. Thus, we then obtain

$$\hat{f}_t = \bar{r}_t^* \quad (4)$$

The resulting estimator of β is given by

$$\hat{\beta} = \left(\sum_{i=1}^N (x_{i,-1}^*)' M_f x_{i,-1}^{**} \right)^{-1} \sum_{i=1}^N (x_{i,-1}^{**})' M_f r_i^* \quad (5)$$

Where $x_{i,-1}^* = (x_{i,1}^*, \dots, x_{i,T-1}^*)'$, $x_{i,-1}^{**} = (x_{i,1}^{**}, \dots, x_{i,T-1}^{**})'$ are $(T - 1) \times m$, and $r_i^* = (r_{i,2}^*, \dots, r_{i,T}^*)'$ and $\hat{f} = (\hat{f}_2, \dots, \hat{f}_T)'$ are $(T - 1) \times 1$.

This model is an extension of the typical predictive regression model used in time-series modeling. The novelty of this model is that the variable $x_{i,t}$ can be treated as a black box, in the sense that the predictor can be stationary or nonstationary (can contain a unit root). Additionally, the predictors are not restricted to being cross-sectionally independent. For further details on the model, see Westerlund et al. (2017).

4. Empirical findings

This section has two objectives. First, we report selected results of the preliminary analysis of the data set, from both time-series and country panel perspectives. Second, we test for bond excess return predictability and discuss the results based on time-series and panel predictive regression models.

4.1. Preliminary analysis

We begin by reporting selected descriptive statistics for our dependent variable (bond excess returns) and five commodity-based predictor variables (*GFP*, *GP*, *OFF*, *OP*, and *WCPI*).² The plots of the data are available on an online Appendix.

We now turn to the selected descriptive statistics of bond excess returns and the five commodity variables. More specifically, we consider the mean, standard deviation, skewness, and test results of the Jarque–Bera (JB) test, which examines the null hypothesis of normality. These results are not tabulated here but are available in an online Appendix. In annualized terms, the mean bond excess return ranges between 2.4% (Norway) and 5.11% (Ireland), and its volatility ranges between 12.16% for Japan and 45.6% for Portugal. Bond excess returns show substantial deviation from normality, as the JB test rejects the null hypothesis of normal distribution for 14 out of the 25 OECD countries at the 5% level of significance. We do not reject the null of normality for the following 11 countries: Australia, Austria, Canada, Denmark, Finland, France, Germany, the Netherlands, Norway, Sweden, and Switzerland.

The mean prices for all five commodities are positive and range from 4.03 (*OFF* and *OP*) to 6.6 (*GP* and *GFP*). We also note that *GFP* and *GP* are the most volatile commodity predictors. Additionally, we note that the null of the JB test is rejected at the 1% level of statistical significance, since none of the five commodity price series follow a normal distribution.

Next, we examine the three salient features of our predictor variables: persistency, endogeneity, and heteroscedasticity. In untabulated results, the AR(1) coefficient is close to one for all five commodity variables, which indicates that all the commodity series (*GP*, *GFP*, *OP*, *OFF*, and *WCPI*) are highly persistent. The ADF unit root test suggests that we are unable to reject the null hypothesis of a unit root in any of the five commodity predictors. The ARCH test results suggest that we do not reject the null of no ARCH for any of the five commodity predictors. Therefore, we conclude that the five commodity variables considered are highly persistent and follow a nonstationary process. Finally, we do not find an ARCH effect in the predictability models when we consider commodities as predictor variables.

Now we examine the same AR(1) model, the ADF unit root test, and the Lagrange multiplier ARCH effect test for bond excess returns and the remaining seven predictor variables. Overall, we conclude that the three salient features of data—persistency, endogeneity, and heteroscedasticity—depend on the choice of predictor variable and on the country considered for the predictability analysis.³ This evidence is sufficient for us to conclude that persistency, endogeneity, and heteroskedasticity need to be accounted for in the predictive regression model.

In addition, since our predictability analysis is based on both time-series and panel data models, it is equally essential to conduct a preliminary analysis of the data in panel form. Thus, we consider evidence on cross-sectional dependence in seven non-commodity predictor variables and bond excess returns. In particular, we draw on the average pairwise cross-sectional correlation coefficients of all eight variables, including the dependent variable. The correlation coefficients fall in the range [−0.039, 0.122]. The null hypothesis of no cross-sectional correlation is rejected for only two predictors, *CPI* and *IP*, at the 5% level of significance.

Finally, we report the panel unit root test proposed by Choi (2001). The null hypothesis of a panel unit root is rejected at the 5% level of significance for three (*CPI*, *DY*, and *IP*) of the 12 predictor variables. For the remaining nine predictors, we do not reject the null of panel unit root. This finding indicates that three of the 12 predictor variables follow a stationary process, whereas the remaining nine are nonstationary variables in panel form. Detailed results are available in an online appendix.

4.2. Predictability test results, using a time-series model

In this section, we discuss the predictability test results using time-series models. First, we use WN's feasible quasi-generalized least squares–based estimators to examine the in-sample predictability of bond excess returns. Specifically, Table 2 reports the in-sample predictability test results for 25 OECD countries. We consider five commodity predictors and seven non-commodity (macroeconomic and financial) variables. We summarize the results as follows. First, we find that, for 12 countries (Canada, the Czech Republic, Denmark, Hungary, Italy, Japan, Mexico, New Zealand, Norway, Poland, Sweden, and Switzerland), there is no evidence of bond excess return predictability. In other words, none of the 12 predictor variables are statistically significant for these 12 countries. For the remaining 13 countries, we find that at least one of the 12 predictor variables significantly predicts bond excess returns. We find that *TS* is the most effective predictor, since it predicts bond excess returns for seven countries (Australia, Austria, Belgium, Finland, Ireland, Portugal, and Spain). Our results are consistent with those of Fama and French (1989), who also find that *TS* is a successful predictor of bond excess returns. Next, we note that *DY* is the second most effective predictor, since it is statistically significant for four countries, Austria, France, Germany, and the UK. Finally, we note that six predictors, *WCPI*, *GP*, *OFF*, *OP*, *CPI*, and *ER*, provide no evidence of statistically significant bond excess return predictability.

Next, we consider five well-known OOS forecast evaluation statistics: Clark and McCracken's (2001) forecast-encompassing test (*ENC* – *NEW*) along with boot-strapped *p*-values, Campbell and Thompson's (2008) OOS *R*-squared (*OOSR2*), relative root mean-squared error (*RRMSE*), the relative mean absolute error (*RMAE*), and the relative Theil's *U* statistics (*RTU*). We obtain the initial forecast value by using the first 50% of the data from the sample. Then, we consider the first

² Due to space constraints, we do not tabulate the preliminary test results for the remaining seven predictor variables (i.e., *CPI*, *DY*, *ER*, *IP*, *Stock*, *TS*, and *T-bill*) in this paper. However, these statistics are available upon request from the authors.

³ . These detailed tabulated results are available upon request from the authors.

Table 2

Time-series in-sample predictability results for OECD countries. This table reports results from the time-series predictive regression model proposed by WN (2015). The regression model regresses bond excess returns on the one-period lagged predictor variable. The null hypothesis is that the predictor variable does not predict bond excess returns. For each country's predictive regression model, we report the coefficient on the one-period lagged predictor variable and its corresponding *p*-value in the parenthesis. ***, **, and * indicate statistical significance at 1%, 5%, and 10% levels, respectively.

	GFP	GP	OPF	OP	WCPI	CPI	DY	ER	IP	Stock	TS	T-bill
Country	β	β	β	β	β	β	β	β	β	β	β	β
Australia	0.001 (0.613)	0.001 (0.617)	0.002 (0.546)	0.002 (0.547)	0.001 (0.633)	0.002 (0.841)	0.006 (0.451)	0.001 (0.915)	–	–0.001 (0.832)	–0.002* (0.068)	0.001 (0.330)
Austria	0.001 (0.637)	0.001 (0.638)	0.000 (0.909)	0.000 (0.891)	0.000 (0.963)	0.004 (0.724)	0.008** (0.014)	–0.002 (0.970)	–0.001** (0.013)	–0.006** (0.045)	0.002** (0.013)	–0.001* (0.056)
Belgium	0.001 (0.585)	0.001 (0.586)	0.000 (0.827)	0.000 (0.808)	0.000 (0.950)	0.004 (0.606)	0.001 (0.580)	–0.003 (0.845)	0.000 (0.405)	–0.004 (0.372)	0.002** (0.046)	–0.001 (0.249)
Canada	0.000 (0.964)	0.000 (0.963)	0.000 (0.839)	0.000 (0.835)	–0.001 (0.741)	–0.007 (0.580)	0.001 (0.889)	–0.001 (0.864)	–0.002 (0.242)	–0.002 (0.501)	0.000 (0.550)	0.000 (0.992)
Czech Republic	0.001 (0.667)	0.001 (0.669)	0.001 (0.854)	0.001 (0.846)	0.000 (0.893)	0.004 (0.795)	0.001 (0.858)	0.004 (0.740)	–	–0.002 (0.367)		0.000 (0.877)
Denmark	0.001 (0.625)	0.001 (0.625)	0.000 (0.853)	0.000 (0.855)	–0.001 (0.760)	–	0.001 (0.839)	0.009 (0.811)	0.000 (0.817)	–0.001 (0.609)	0.002 (0.843)	0.000 (0.864)
Finland	0.001 (0.636)	0.001 (0.636)	0.000 (0.994)	0.000 (0.975)	0.000 (0.86)	0.006 (0.688)	0.006 (0.118)	0.012 (0.744)	–	–0.011** (0.016)	0.002** (0.011)	–0.001* (0.056)
France	0.002 (0.591)	0.002 (0.589)	0.001 (0.788)	0.001 (0.773)	0.000 (0.907)	0.002 (0.910)	0.009* (0.064)	0.004 (0.922)	–0.001 (0.961)	–0.004 (0.365)	0.001 (0.138)	–0.001 (0.445)
Germany	0.001 (0.652)	0.001 (0.653)	0.000 (0.853)	0.000 (0.841)	0.000 (0.978)	0.005 (0.739)	0.01* (0.053)	0.003 (0.913)	–0.006 (0.791)	–0.002 (0.59)	0.001 (0.607)	0.000 (0.702)
Hungary	0.002 (0.294)	0.002 (0.294)	0.000 (0.446)	0.000 (0.445)	0.000 (0.453)	0.005 (0.206)	0.004 (0.656)	–0.041 (0.293)	–	0.001 (0.784)	0.001 (0.319)	–0.001 (0.425)
Ireland	0.004 (0.255)	0.004 (0.255)	0.003 (0.505)	0.003 (0.480)	0.004 (0.468)	0.011 (0.637)	–0.007 (0.117)	–0.007 (0.388)	0.000 (0.723)	–0.005 (0.404)	0.001** (0.017)	–0.002 (0.133)
Italy	0.000 (0.622)	0.000 (0.626)	–0.001 (0.895)	–0.001 (0.892)	–0.001 (0.967)	–0.003 (0.683)	0.001 (0.984)	–0.012 (0.719)	–0.005 (0.639)	–0.014 (0.145)	0.001 (0.252)	–0.001 (0.370)
Japan	0.000 (0.901)	0.000 (0.900)	0.000 (0.989)	0.000 (0.996)	0.000 (0.953)	0.036 (0.410)	0.002 (0.494)	0.004 (0.557)	0.041 (0.313)	–0.001 (0.804)	–0.002 (0.325)	0.003 (0.331)
Korea	0.000 (0.833)	0.000 (0.833)	–0.002 (0.379)	–0.002 (0.350)	–0.003 (0.349)	–0.004 (0.675)	0.002 (0.669)	–0.011 (0.346)	–0.002*** (0.001)	–0.003 (0.285)	–	0.000 (0.983)
Mexico	–0.002 (0.495)	–0.002 (0.497)	–0.004 (0.298)	–0.004 (0.285)	–0.004 (0.286)	–	0.007 (0.434)	0.011 (0.210)	0.002 (0.222)	–0.002 (0.242)	–	0.000 (0.990)
Netherlands	0.001 (0.606)	0.001 (0.607)	0.000 (0.941)	0.000 (0.920)	0.000 (0.931)	0.006 (0.694)	0.004 (0.364)	0.002 (0.962)	0.000 (0.647)	–0.01** (0.046)	0.002 (0.348)	0.000 (0.518)
New Zealand	0.002 (0.398)	0.002 (0.400)	0.001 (0.672)	0.001 (0.689)	0.001 (0.668)	0.005 (0.624)	0.000 (0.930)	0.001 (0.726)	–	–0.001 (0.607)	0.001 (0.600)	0.000 (0.739)
Norway	0.001 (0.461)	0.001 (0.462)	0.001 (0.571)	0.001 (0.574)	0.001 (0.648)	0.007 (0.566)	0.005 (0.170)	0.002 (0.895)	0.001 (0.172)	0.000 (0.828)	0.000 (0.907)	0.000 (0.768)
Poland	0.000 (0.938)	0.000 (0.938)	–0.001 (0.768)	–0.001 (0.762)	–0.001 (0.769)	0.000 (0.797)	0.001 (0.624)	0.009 (0.358)	–	–0.005 (0.252)	–0.003 (0.109)	0.000 (0.403)
Portugal	0.001 (0.590)	0.001 (0.597)	0.000 (0.572)	0.000 (0.564)	–0.001 (0.606)	–0.006 (0.676)	0.006 (0.181)	0.019 (0.826)	0.000 (0.345)	–0.014 (0.128)	0.001* (0.060)	–0.002 (0.292)
Spain	0.001 (0.322)	0.001 (0.320)	0.000 (0.593)	0.000 (0.584)	0.000 (0.612)	0.005 (0.504)	0.002 (0.353)	0.002 (0.808)	–	–0.002 (0.515)	0.00** (0.047)	–0.002** (0.047)
Sweden	0.000 (0.812)	0.000 (0.811)	0.000 (0.856)	0.000 (0.839)	0.000 (0.999)	0.001 (0.881)	0.003 (0.200)	0.000 (0.996)	–0.001 (0.548)	–0.003 (0.380)	–0.001 (0.449)	0.000 (0.785)
Switzerland	0.000 (0.840)	0.000 (0.845)	–0.001 (0.686)	–0.001 (0.694)	–0.002 (0.629)	0.001 (0.957)	0.002 (0.434)	0.004 (0.53)	–	–0.003 (0.345)	0.001 (0.466)	0.000 (0.582)
U.K.	0.000*** (0.000)	0.003 (0.267)	0.002 (0.582)	0.002 (0.576)	0.002 (0.664)	0.010 (0.310)	0.017* (0.099)	–0.012 (0.264)	0.025 (0.422)	0.001 (0.718)	0.000 (0.305)	–0.001 (0.241)
U.S.	0.000*** (0.000)	0.000 (0.957)	–0.001 (0.869)	0.000 (0.875)	–0.001 (0.767)	–0.012 (0.415)	0.001 (0.766)	0.001 (0.968)	0.192 (0.506)	–0.003 (0.624)	0.000 (0.776)	0.000 (0.922)

50% of the data plus one month of observations and generate the second forecast value. Similarly, we add one observation at a time, until the entire sample is considered to obtain the forecast returns. In other words, our initial 50% in-sample period (January 2000 to June 2008) is used to generate recursive forecasts of bond excess returns for the remaining 50% of the sample. The results can be interpreted as follows. The *ENC* – *NEW* test statistic examines whether restricted model forecasts encompass the unrestricted model forecasts. In other words, this statistic examines whether our proposed model provides no useful information for predicting bond excess returns relative to a constant model of bond excess returns. The *OOSR2* statistic examines the difference in the mean-squared errors from the competitive model and the constant bond excess returns model. If *OOSR2* > 0, our proposed predictability model outperforms the constant returns model. Finally, if *RRMSE*, *RMAE*, and the *RTU* statistics are less than one, our proposed predictive regression model outperforms the constant bond excess returns model.

Since our paper's focus is on panel predictability model, we do not tabulate the time-series-based OOS test results, due to space constraints. However, we briefly summarize our main findings, as follows. After considering the OOS forecast evaluations, the main finding from the time-series predictability analysis is that the evidence of predictability from OOS analysis is much better than the evidence found using an in-sample predictability test. Overall, we can conclude the following from our time-series-based OOS forecast evaluation:

- Models based on *GFP* and *GP* turn out to be the most effective predictability models, compared with the constant returns model. At least one of the five OOS test evaluations reveals that the *GFP*- and *GP*-based predictive regression models beat the historical mean model in 23 of 25 countries.
- The next most effective model considers *CPI* a predictor of bond excess returns. At least one of the five OOS test evaluations favors the *CPI*-based predictability model for 21 of 23 countries.
- The *OPF* is the weakest predictor. It outperforms the constant returns model only for seven countries (Australia, Canada, Hungary, Ireland, Italy, Norway, and the US). Additionally, *OP*- and *DY*-based predictive regression models provide weak evidence in favor of bond excess return predictability, compared with the constant returns model. In particular, *OP*- and *DY*-based predictability models fail to beat the constant returns model in at least 10 of 25 countries.
- For two countries, Australia and Ireland, we find that 11 of 12 (except *IP* and *ER*, respectively) predictor-based models outperform the constant returns model.
- Three countries, the Czech Republic, Germany, and Switzerland, show limited evidence of OOS predictability. For these three countries, only six of the 12 predictor-based models outperform the constant returns model.
- Finally, for nine of 25 countries, we find that four out of five OOS test evaluations favor our *T-bill*-based predictability model over the constant returns model. This finding suggests that the *T-bill* is one of the strongest predictors of bond excess returns, particularly for the following nine countries: Austria, Belgium, Finland, France, Ireland, Japan, the Netherlands, Portugal, and Spain.

4.3. Predictability test results using panel data models

This section presents the results based on the panel predictability model. Our aim in this section is to determine whether the panel predictability model provides better evidence of predictability compared to time-series analysis. In Table 3, we report the results based on the in-sample panel predictability model proposed by Westerlund et al. (2017). We consider 12 predictor variables and a panel of 25 OECD countries; see Table 4 for details. We find that the null hypothesis of no predictability is rejected at the 5% level of significance for five predictor-based panel predictability models. Note also that three of five statistically significant predictors are commodity variables (*OP*, *OPF*, and *WCPI*). The remaining two statistically significant predictors of the bond excess returns of our panel of 25 OECD countries are *TS* and *T-bill*. Additionally, we find no significant evidence of panel bond excess return predictability when we consider the remaining seven predictors (*GFP*, *GP*, *CPI*, *DY*, *ER*, *IP*, and *Stock*).

Next, we conduct the panel OOS predictability analysis. The panel OOS predictability approach is the same as that of the time-series OOS evaluation. More specifically, a 50% in-sample period (January 2000 to June 2008) is used to generate recursive forecasts of bond excess returns for the remaining 50% of the sample. To compare the performance of our 12 predictor-based panel predictability regression models, we specify a benchmark constant returns model. Similarly, we examine panel OOS predictability for a constant bond excess returns model. The panel OOS predictability evidence is based on the following five evaluations: *OOSR2*, *RTU*, *RRMSE*, *RMAE*, and the *ENC – NEW* forecast-encompassing test statistics. The results are reported in Table 5.⁴

First, the coefficients for the *RTU*, *RRMSE*, and *RMAE* test statistics for all 12 panel predictability models are less than one. This result indicates that a forecasting model that uses our 12 proposed predictors to forecast bond excess returns beats the constant returns model. Second, for the panel predictability results based on the null hypothesis $OOSR2 \leq 0$, the null is rejected in the case of all 12 panel predictability models and the coefficients are positive. This results again suggests that our proposed predictability models perform better than the benchmark constant returns model. Finally, we consider the results for the *ENC – NEW* test statistics. The *p*-values for the *ENC – NEW* statistics are reported in the final column. Note that all *p*-values are greater than 0.1, which means the *ENC – NEW* statistics are statistically insignificant. This means that we do not reject the null that our proposed panel predictability models do not contain enough information to forecast bond excess returns. Thus, we conclude that four out of the five OOS test evaluations provide significant evidence that our proposed panel predictability models outperform the historical mean model.

We consider 12 predictor variables in our empirical analysis. Among them, five are the commodity variables *GP*, *GFP*, *OP*, *OPF*, and *WCPI*, and seven are non-commodity variables (five macroeconomic and two financial variables), namely, *CPI*, *DY*, *ER*, *IP*, *Stock*, *TS*, and *T-bill*. We summarize our findings from the two predictive regression methods, time-series and panel regressions. Most importantly, the panel predictability models provide more significant evidence compared to time-series regressions, which is particularly true in OOS evaluations. Overall, we note that four out of five OOS matrixes provide sig-

⁴ In additional tests, as suggested by one of the referees, we estimate out-of-sample tests using five lags. The results are insensitive to this lag length selection. Detailed results are available upon request.

Table 3

In-sample panel predictability test results. In this table, we report the results from the panel predictive regression model proposed by Westerlund et al (2017). The regression model regresses the bond excess returns of a panel of OECD countries on one-period lagged predictor variable. Predictor variables considered in our study are gold futures price (*GFP*), gold spot price (*GP*), oil futures price (*OF*), oil spot price (*OP*), world commodity index (*WCPI*), consumer price index (*CPI*), dividend yield (*DY*), exchange rate (*ER*), industrial production (*IP*), stock market index (*Stock*), term spread (*TS*), and 3-month Treasury bill yield (*T-bill*). The null hypothesis is that the proposed predictor variable does not predict bond excess returns. *** and ** indicate statistical significance at 1% and 5% significance levels, respectively.

Predictor	β	t-stat	No. of Countries
<i>GFP</i>	−0.000	−0.344	25
<i>GP</i>	−0.000	−0.352	25
<i>OF</i>	0.001***	2.907	25
<i>OP</i>	0.001***	2.917	25
<i>WCPI</i>	0.001**	2.516	25
<i>CPI</i>	0.001	0.600	23
<i>DY</i>	−0.002	−0.944	25
<i>ER</i>	−0.010	−0.434	25
<i>IP</i>	0.000	0.102	17
<i>Stock</i>	−0.047	−0.878	25
<i>TS</i>	−0.001***	−5.911	22
<i>T-bill</i>	0.002***	3.845	25

Table 4

List of predictors for OECD high-income, OECD-European region, and G7 panels. This table shows the predictor availability for OECD panels of high-income (Panel A), European (Panel B), and G7 (Panel C). Column 2 contains the number of countries and the country list is reported in the last column. There are 12 predictors, namely, gold futures price (*GFP*), gold spot price (*GP*), oil futures price (*OF*), oil spot price (*OP*), world commodity index (*WCPI*), consumer price index (*CPI*), dividend yield (*DY*), exchange rate (*ER*), industrial production (*IP*), stock market index (*Stock*), term spread (*TS*), and 3-month Treasury bill yield (*T-bill*).

Predictor	No. of Countries	Country List
Panel A: OECD high-income panel <i>GFP, GP, OF, OP, WCPI, ER, DY, T-bill, Stock</i>	23	Australia, Austria, Belgium, Canada, Czech Republic, Denmark, Finland, France, Germany, Hungary, Ireland, Italy, Japan, Korea, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, U.K., and U.S.
<i>CPI</i>	22	Australia, Austria, Belgium, Canada, Czech Republic, Finland, France, Germany, Hungary, Ireland, Italy, Japan, Korea, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, U.K., and U.S.
<i>IP</i>	16	Austria, Belgium, Canada, Denmark, France, Germany, Ireland, Italy, Japan, Korea, Netherlands, Norway, Portugal, Sweden, U.K., and U.S.
<i>TS</i>	21	Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Hungary, Ireland, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, U.K., and U.S.
Panel B: OECD-European region panel <i>GF, GFP, OF, OFP, WCPI, ER, DY, T-bill, Stock</i>	18	Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Hungary, Ireland, Italy, Netherlands, Norway, Poland, Portugal, Spain, Sweden, Switzerland, and U.K.
<i>CPI</i>	17	Austria, Belgium, Czech Republic, Finland, France, Germany, Hungary, Ireland, Italy, Netherlands, Norway, Poland, Portugal, Spain, Sweden, Switzerland, and U.K.
<i>IP</i>	11	Austria, Belgium, Denmark, France, Germany, Italy, Netherlands, Norway, Portugal, Sweden, and U.K.
<i>TS</i>	17	Austria, Belgium, Denmark, Finland, France, Germany, Hungary, Ireland, Italy, Netherlands, Norway, Poland, Portugal, Spain, Sweden, Switzerland, and U.K.
Panel C: G7 country panel <i>GF, GFP, OF, OFP, WCPI, ER, DY, T-bill, Stock, CPI, IP, TS</i>	7	Canada, France, Germany, Italy, Japan, U.K., and U.S.

nificant evidence in favor of our proposed 12 predictor-based models, over the benchmark constant bond excess returns model. Moreover, we note strong evidence in favor of commodity-based predictability models, compared to non-commodity-based predictability models. In particular, three of the five commodity variables (*OP*, *OF*, and *WCPI*) provide consistent and significant evidence of bond excess return predictability, whereas only two of the seven non-commodity variables (*TS* and *T-bill*) are found to be significant predictors of bond excess returns.

5. Economic significance

Statistical forecasting performance measures are sometimes found to be insufficient from the practitioner's perspective. In this regard, it is therefore important to provide evidence on the economic significance of our proposed forecasting models. In other words, one could consider how investors can successfully trade by tracking information from predictors found to be significant predictors of bond excess returns. We address this by utilizing forecast bond excess returns from our predictor-

Table 5

Panel out-of-sample predictability test results. This table reports out-of-sample predictability test results for a panel of OECD countries. More specifically, here we report *RTU*, *RRMSE*, *RMAE*, *OOSR2*, and *ENC – NEW* statistics. *RTU*, *RRMSE*, and *RMAE* values less than one and $> \text{OOSR2}$ indicate that our proposed predictive regression model outperforms historical mean model. The *ENC – NEW* test statistic examines whether restricted model forecasts encompass the unrestricted model forecasts. We report *ENC – NEW* statistics along with the bootstrapped *p*-values in last column.

Predictor	Theil's <i>U</i>			<i>RRMSE</i>	<i>RMAE</i>	<i>OOSR2</i>	<i>ENC – NEW</i>	
	Model	Constant	<i>RTU</i>				Statistic	<i>p</i> -value
<i>GFP</i>	0.643	0.863	0.745	0.923	0.861	0.147	1.999	0.280
<i>GP</i>	0.643	0.863	0.745	0.923	0.861	0.147	1.990	0.270
<i>OFP</i>	0.644	0.863	0.746	0.922	0.860	0.148	1.615	0.270
<i>OP</i>	0.644	0.863	0.746	0.922	0.860	0.148	1.610	0.259
<i>WCPI</i>	0.644	0.863	0.746	0.922	0.860	0.148	1.281	0.184
<i>CPI</i>	0.652	0.863	0.755	0.923	0.866	0.137	0.293	0.228
<i>DY</i>	0.629	0.863	0.730	0.913	0.839	0.149	1.465	0.325
<i>ER</i>	0.645	0.863	0.748	0.922	0.859	0.148	0.176	0.279
<i>IP</i>	0.644	0.941	0.684	0.913	0.838	0.152	0.201	0.500
<i>Stock</i>	0.646	0.863	0.748	0.920	0.854	0.139	−0.529	0.279
<i>TS</i>	0.633	0.942	0.672	0.907	0.826	0.167	0.509	0.201
<i>T-bill</i>	0.644	0.863	0.746	0.923	0.861	0.147	−0.239	0.400

based models to construct a power utility function. In particular, we closely follow [Gargano et al. \(2017\)](#) and [Sarno, Schneider, and Wagner \(2016\)](#). Utility gains are simply the difference between the utility of our proposed model forecast and the utility of historical average forecasts. We conduct a power utility function for both time-series and panel predictability-based forecasts. The economic significance analysis is conducted only for those models where we find evidence of significant predictability, irrespective of the use of in-sample and OOS test evaluations. For instance, if any variable is found to be a insignificant predictor of bond excess returns, we do not analyze that particular predictability model's economic significance.⁵ Note also that we use 50% of the sample as an in-sample period and generate recursive forecasts of bond excess returns for the remaining 50% of the sample.

Utility gains are calculated as the difference in utility from two models, our proposed time-series, the predictor-based predictability model, and the constant bond excess returns model. The utility function is as follows:

$$U(W_{t+1}) = W_{t+1}^{1-\rho} \quad (6)$$

$$W_{t+1} = W_t \times R_{t+1} \quad (7)$$

$$R_{t+1} = 1 + r_f + w_t \cdot r_{t+1} \quad (8)$$

where r_f is the risk-free rate, r_{t+1} is the bond excess return, W_t is wealth, and w_t is the optimal weight. We assume initial wealth of \$1. The portfolio contains the riskless bond and the risky bond with conditional variance σ_t^2 , where the risk aversion factor is equal to three ($\rho = 3$). The weight is restricted to the range from zero and one, $w(0, 1)$, implying no short selling or borrowing. The weight is computed as follows:

$$w_t = \frac{E(r_{t+1}) + \frac{1}{2}\sigma_t^2}{\rho\sigma_t^2} \quad (9)$$

The results based on time-series utility gains are reported in [Table 6](#). Positive (greater than or equal to zero) utility gains indicate that our proposed model outperforms the benchmark constant bond excess returns model. More specifically, our results can be summarized as follows:

- We find the positive utility gains of the *GFP* predictor to be highest, compared with those obtained from the other 11 predictors. The *GFP*-based utility gains are positive for 11 of 23 countries (Australia, Denmark, Hungary, Ireland, Italy, Korea, Mexico, New Zealand, Poland, Portugal, and Spain).
- The second most economically successful predictors are *GP*, *CPI*, *TS*, and *T-bill*. The utility gains obtained using these four predictors are positive in the case of 17 countries. Additionally, note that *GP*, *CPI*, *TS*, and *T-bill* based utility gains are reported as negative for six countries (Austria, Belgium, the Czech Republic, Ireland, the Netherlands, and the US), four countries (Belgium, the Czech Republic, Ireland, and Portugal), four countries (Australia, Ireland, Italy, and Sweden), and five countries (Germany, Ireland, Italy, Norway, and Portugal), respectively.
- Additionally, note that, for five predictors, *OP*, *WCPI*, *DY*, *ER*, and *Stock*, the utility gains are reported as positive for at least 10 countries.

⁵ It is important to note that, OOS, panel predictability evaluation provides strong evidence in favor of our proposed panel predictability model over the constant bond excess returns model. We therefore utilize all 12 predictor-based models to compute utility gains.

Table 6

Time-series country specific utility gains. This table reports utility gains which is defined as utility from our proposed predictability model minus the utility from the benchmark constant bond excess returns model. We use Sarno et al. (2016) power utility function which is represented as follows: $U(W_{t+1}) = W_{t+1}^{1-\rho}$ and, $W_{t+1} = W_t \times R_{t+1}$ and $R_{t+1} = 1 + r_f + w_t \cdot r_{t+1}$. Here r_f is the risk-free rate, r_{t+1} is the bond excess returns, w_t is the optimal weight. The weight is given by $w_t = \frac{E(r_{t+1}) + A/\sigma_t^2}{\rho \sigma_t^2}$. Weights lie between 0 and 1, in which short-selling and borrowing are not allowed. We assume initial wealth is \$1. The portfolio contains the riskless bond and risky bond with conditional variance σ_t^2 with the risk aversion, $\rho = 3$. There are 12 predictors, namely, gold futures price (GFP), gold spot price (GP), oil futures price (OPF), oil spot price (OP), world commodity index (WCPI), consumer price index (CPI), dividend yield (DY), exchange rate (ER), industrial production (IP), stock market index (Stock), term spread (TS), and 3-month Treasury bill rate (T-bill). The results are for the countries that have statistically significant in-sample and/or out-of-sample predictability only.

Country	GFP	GP	OPF	OP	WCPI	CPI	DY	ER	IP	Stock	TS	T-bill
Australia	0.715	0.766	0.000	0.000	0.001	0.692	-0.015	-0.761	-	-2.847	-0.187	5.266
Austria	-0.003	-0.002	-	-	0.000	0.000	0.000	-	0.111	-	0.433	0.000
Belgium	-0.048	-0.055	-	-	-0.023	-0.007	-0.004	-	0.380	-0.143	0.001	0.000
Canada	-	-	0.000	0.000	0.000	-	-	0.000	0.018	-	0.000	0.000
Czech Republic	-0.033	-0.033	-0.001	-0.001	-0.001	-0.014	0.000	-	-	-	-	-
Denmark	0.083	0.083	-	0.000	-0.004	-	-	-	0.000	0.013	0.420	0.040
Finland	0.000	0.000	-	-	0.000	0.000	-	0.000	-	0.000	0.054	0.000
France	0.000	0.000	-	-	-	0.000	0.000	-	0.001	0.000	0.000	0.000
Germany	0.000	0.000	-	-	0.000	0.000	-	-	-3.782	-	0.000	-0.004
Hungary	13.609	13.606	13.165	13.109	13.618	13.102	13.621	13.223	-	13.698	-	58.620
Ireland	0.005	-0.131	-0.063	0.039	-0.009	-1.407	0.033	-	-0.007	0.000	-0.020	-0.017
Italy	0.180	0.176	-0.769	-0.500	0.197	0.011	-	-	-0.003	0.000	-0.003	-0.009
Japan	-	-	-	-	-	0.193	0.000	0.000	0.000	0.000	0.684	0.006
Korea	0.147	0.147	-	-	-	0.190	0.128	0.000	1.022	0.242	-	0.078
Mexico	0.469	0.469	-	-	-	-	-	0.476	4.691	0.472	-	0.000
Netherlands	-0.008	-0.007	-	-	-0.004	0.000	-	-	-	0.000	0.000	0.000
New Zealand	2.946	2.946	-	0.000	0.000	2.805	-	0.000	-	4.036	2.497	-
Norway	0.000	0.000	0.000	0.000	0.000	0.000	3.475	0.000	3.154	-6.999	0.000	-0.003
Poland	1.035	1.035	0.398	0.588	0.631	0.000	0.000	0.000	-	1.211	0.000	0.000
Portugal	0.219	0.220	-	-0.136	-2.189	-2.454	-	-0.031	-	-0.006	0.142	-0.026
Spain	0.652	0.650	-	0.792	0.658	0.026	-	0.000	-	-	0.000	0.000
Sweden	0.000	0.000	-	-	0.000	0.000	-	-0.006	-0.010	-	-0.002	-
Switzerland	0.000	0.000	-	-	-	0.000	-	0.000	-	-	0.003	0.000
U.K.	0.000	0.000	-	0.000	0.000	0.000	0.015	-0.605	-	0.031	0.000	0.061
U.S.	-0.002	-0.001	0.000	0.000	-	-	0.012	-	-1.804	-0.002	0.000	0.000

- For the predictors *OPF* and *IP*, we find that utility gains are positive only in the case of two countries (Hungary and Poland) and seven countries (Austria, Belgium, Canada, France, Korea, Mexico, and Norway), respectively.

Next, we consider the utility gains for the OECD panel. We report the results in Table 7. Note that, when we consider all the OECD countries in a panel, we find that the utility gains for all 12 predictor variables are positive. This finding implies that, for the OECD panel, our proposed predictor-based models beat the constant bond excess returns model, irrespective of the predictor considered in forecasting panel bond excess returns. Thus, we conclude from our economic significance analysis that all 12 predictors are economically significant when all 25 OECD countries are considered in a panel. Note that our results are contrary to those reported by Thornton and Valente (2012). These authors fail to identify statistical evidence of predictability, which translates into economic value when compared with the benchmark model. However, our findings are consistent with those of Gargano et al. (2017), who find economic gains from their predictability models. They authors conduct predictability analyses of US Treasury bonds with maturities ranging from two to five years. The predictors used in their study are the forward spread, a linear combination of forward rates, and the Ludvigson and Ng (2009) macro factors.

6. Robustness tests

In this section, we undertake a robustness test analysis. As a robustness check, from the panel of 25 OECD countries, we form three more panels: OECD high-income countries, OECD European region countries, and G7 countries.⁶ We examine whether our obtained panel predictability results are not biased toward the particular OECD panel. To do so, we construct three additional panels: the OECD high-income country panel, the OECD European region country panel, and the G7 country panel. For the sake of completeness, in this section, we repeat the panel predictability analysis for the three newly constructed panels. Our results are as follows with detailed results available in an online appendix:

⁶ We also check whether our results on the time-series predictability of bond excess returns are robust to different data sample periods. More specifically, we examine the in-sample and OOS predictability evaluations by considering our data set over the period January 1920 to December 2016. These results are reported in an online Appendix.

Table 7

Utility gains of a panel of OECD countries. This table reports utility gains which is defined as utility from our proposed panel predictability model minus the utility from the benchmark constant bond excess returns model for a panel of OECD countries. We use Sarno et al. (2016) power utility function: $U(W_{t+1}) = W_{t+1}^{1-\rho}$ and, $W_{t+1} = W_t \times R_{t+1}$ and $R_{t+1} = 1 + r_f + w_t \cdot r_{t+1}$. Here r_f is the risk-free rate, r_{t+1} is the bond excess returns, w_t is the optimal weight. The weight is given by $w_t = \frac{E(r_{t+1}) + \frac{1}{2}\rho\sigma_t^2}{\rho\sigma_t^2}$. Weights lie between 0 and 1, in which short selling and borrowing are not allowed. We assume initial wealth is \$1. The portfolio contains the riskless bond and risky bond with conditional variance σ_t^2 with the risk aversion, $\rho = 3$.

Predictor	Utility Gain
GFP	0.799
GP	0.795
OFFP	0.517
OP	0.575
WCPI	0.537
CPI	0.571
DY	0.695
ER	0.491
IP	1.342
Stock	0.388
TS	0.779
T-bill	2.674

- Westerlund, Karabiyik, and Narayan's (2017) in-sample panel predictability test reveals that five of the 12 predictors, namely, *OFFP*, *OP*, *WCPI*, *TS*, and *T-bill*, significantly predict the bond excess returns of the OECD high-income and OECD European region panels. Our results for these two are consistent with our main results (i.e., when we consider all 25 OECD countries in a panel).
- For the G7 panel, we find that only three variables, *OFFP*, *OP*, and *TS*, are statistically significant predictors of bond excess returns. The variables *WCPI* and *T-bill* fail to significantly predict G7 bond excess returns but are found to be significant predictors for the other OECD-based panels, as discussed above.
- For four out of five OOS test evaluations, *RTU*, *RRMSE*, *RMAE*, and *OOSR2* provide evidence in favor of our proposed predictability model (irrespective of the predictors considered) over the benchmark constant bond excess returns model for all three panels, that is, for the OECD high-income, OECD European region, and G7 countries. Based on the *ENC – NEW* OOS test evaluation, we find no evidence in favor of our proposed predictability model over the constant bond excess returns model for any of the three OECD-based panels.

Overall, we conclude that our in-sample and OOS predictability tests provide robust evidence on panel predictability, irrespective of the panel considered. More specifically, based on an in-sample predictability test, five of 12 predictors significantly predict bond excess returns in the majority of the panels. Moreover, OOS predictability test evaluation provides strong evidence in favor of all 12 predictor-based models over the constant bond excess returns model, irrespective of the OECD panel considered. However, we do not find robust results when we consider the *ENC – NEW* OOS test evaluation.

7. Conclusion

In this paper, we examine the bond excess return predictability for 25 OECD countries. Our main data set is monthly, covering the period from January 1920 to December 2016. We employ two methods, time-series and panel predictive regression models, to forecast bond excess returns. We conclude two key findings. Our first and main finding is that three commodity variables (*OFFP*, *OP*, and *WCPI*) predict bond excess returns. This is a new finding for the bond predictability literature. Specifically, we perform the analysis with four different panel formations. The two commodity predictors *OFFP* and *OP* show significance in all four panel formations, namely, the OECD, OECD high-income, OECD European region, and G7 country panels. Thus, among the five commodity variables, we find that these three predict bond excess returns.

Our second finding relates to the predictive ability of other macroeconomic and finance variables. The predictor *TS* consistently predicts bond excess returns, regardless of whether we employ time-series or panel data regressions. With regard to investor utility, we find considerable evidence of utility gains from both the time-series and panel-based forecasting models. This result contrasts with the findings of Thornton and Valente (2012), who conclude that statistical evidence of bond return predictability is unsuccessfully translated into economic return predictability. Nonetheless, our economic significance results are consistent with those of Gargano et al. (2017), who show economically significant return predictability in the US Treasury bond market.

From a regulatory and implication points of view our paper has several indirect messages. First, in our story, commodity markets are the most important factor for movements in bond prices. It follows that regulatory bodies, including central banks, would need to monitor commodity price movements. This has become that much more important in light of

COVID-19 which has impacted commodity markets, particularly oil prices (see Narayan, 2020, Devpura and Narayan, 2020).⁷ Second, regulatory issues require forecasting of bond prices. It also often requires forecasting of other financial indicators such as inflation rate for example. Our results show that in forecasting any of these financial indicators, it is important to account for the commodity prices. Doing so is likely to improve forecasts and therefore bring greater precision to regulatory decision making. Third, our model maybe particularly useful for devising trading strategies given that we show significant utility gains do arise for bond market investors. This evidence is however likely to change in light of the COVID-19 pandemic which has been response for market volatility (see Sharma, 2020) and uncertainty (Iyke 2020b). It follows that future studies should address the shock of COVID-19 in estimating utility gains and in forecasting as shown by Devpura (2020), Gil-Alana and Claudio-Quiroga (2020), Appiah-Otoo (2020), Lan, Huang, and Huang (2020), Yan and Qian (2020), Han and Qian (2020), Yang and Deng (2021), Hoang and Syed (2021), Fang and Zhang (2021), Wang and Su (2021), Narayan, Devpura and Wang (2020), Chen, Liu and Zhao (2020), Liu, Choo and Lee (2020), Mishra, Rath, and Dash (2020), Ming, Zhou, Ai, and Zhong (2020), Phan and Narayan (2020), Salisu and Akanni (2020), Shen, Fu, Pan, Yu, and Chen (2020) Song, Zhang, Zhao and Xu (2020), Wang, Zhang, Wang, and Fu (2020), amongst others.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.intfin.2021.101301>.

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⁷ For the effect of COVID-19 and oil relationship, see Gil-Alana and Monge (2020); Liu et al. (2020); Iyke (2020a) Qin et al. (2020); Huang and Zheng (2020); and Prabheesh et al. (2020). Literature summarizes on COVID-19 can be read in Sha and Sharma (2020) and Sharma and Sha (2020).

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